

I design and implement complex software systems: modular .NET platforms, backend services with payments and recovery, compilers, and program-analysis tools. My strongest areas are explicit contracts, state invariants, verifiable semantics, and carrying systems through to reproducible outcomes.

Platform architecture

UniversalToolchain/Wist2: independent packages, typed contracts, routes, manifests/locks, and lifecycle ownership

Production backend

Payments, subscriptions, roles, devices, PostgreSQL/SQLite, migrations, recovery, backup/restore, and rollback

Compiler / systems

Callable-first SSA, interpreter/CIL, LLVM, C++23, memory analysis, and x86-64 code generation

EXPERIENCE

- 2024 — present

UniversalToolchain / Wist2 — creator and primary developer

 - Designed language composition from independently versioned packages: typed artifacts, capabilities and conflicts, deterministic routes, and pass ordering.
 - Implemented exact manifest binding, lock identity, fail-closed policies, and runtime-session lifecycle ownership.
- 2026 — present

.NET backend: VpnMediator and CompilationLabLMS

 - Designed subscription, payment, balance, role, and device state; handled webhooks and repeated operations idempotently.
 - Built migrations, background recovery, audit trails, backup/restore, health gates, Docker, nginx, systemd, and rollback paths.
- July 1 — August 31, 2026

MCST — compiler engineering intern

 - Developing an LLVM optimization pass and implementing tested graph algorithms in C++23.

PROJECTS

- UniversalToolchain / Wist2

1,465/1,465 tests with zero failures across nine packages and clean-consumer projects, plus interpreter/CIL parity and reproducible configuration checks.
- VpnMediator

A production service with controlled state transitions, backup/restore, health gates, and safe rollback; the public case study excludes secrets and user data.
- PS-form Memory Dependence Analyzer

Ranked 1st of 49 with 104/104 tests, with reproducible wrong-answer and time-limit counterexamples.

SKILLS

- Architecture

module boundaries, typed contracts, state machines, manifests/locks, lifecycle ownership, fail-closed design
- .NET / backend

C#/.NET, ASP.NET Core, REST/OpenAPI, PostgreSQL, SQLite, webhooks, transactions, migrations, recovery
- Compilers

Bytecode/AIR, CFG, SSA, dominance, optimizations, interpreter/CIL, LLVM, program analysis
- Systems and verification

C++23, C17, Rust, Python, Linux, CMake, sanitizers, differential/metamorphic testing, exact oracles

EDUCATION

Software Engineering student at HSE University

RECOGNITION

- HSE Vysshaya Proba prize-winner in competitive and industrial programming; regional Russian informatics olympiad prize-winner.
- Moscow / remote